Value Engineering Workshop - Minutes

Cab structure and trim, two-day workshop held 12 and 13 September 2024

Document number	CVOEM/VAVE/VE-WS/2024/017
Revision	Rev 01 - circulated for correction, no corrections received
Issue date	20 September 2024
Minuted by	VAVE Programme Office
Chair	Chief Engineer - Cab Systems
Venue	Engineering Research Centre, Pune - Room 4B
Attendance	17 of 19 invited; attendance list at section 1
Distribution	All attendees; VAVE Committee; absentees

Summary

A two-day value engineering workshop on cab structure and trim generated 34 ideas, of which 9 were screened out during the workshop and 25 were logged for assessment. Four ideas were identified during the workshop as probable duplicates of existing records and were checked before logging; three were confirmed duplicates and one was found to be a carry-over to a different programme. The NVH package discussed on day two is recorded in section 5.

1 Attendance

Name and role	Function	Day 1	Day 2
Chief Engineer - Cab Systems (Chair)	Product Engineering	Y	Y
Principal Engineer - Cab Structure	Product Engineering	Y	Y
Senior Engineer - Trim and Interiors	Product Engineering	Y	Y
Senior Engineer - NVH	Product Engineering	N	Y
Manager - Cost Engineering	Cost Engineering	Y	Y
Engineer - Cost Engineering	Cost Engineering	Y	N
Manager - Supplier Development	Supplier Development	Y	Y
Lead - Manufacturing Engineering, Cab Shop	Manufacturing	Y	Y
Manager - Field Quality	Field Quality	Y	Y
VAVE Programme Office (minutes)	Programme Office	Y	Y
Seven further engineering attendees	Various	Y	Y

Apologies: Manager - Homologation; Senior Engineer - Electrical Installation. Both have been sent these minutes and asked to comment on items 3.4 and 4.2 respectively.

2 Objectives and method

The chair set three objectives: to generate ideas against the cab structure allocation of 42 kg in CVOEM/VAVE/WR-RM/2024/002; to review the trim cost gap identified in the FY23 teardown; and to

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close out four long-standing proposals that had been deferred twice.

The function-analysis method was used for the structure session and a cost-driver walk for the trim session. Ideas were captured on the board, screened by the group against three criteria, and logged by the programme office at the end of each session.

2.1 Screening criteria applied in the room

- Does it change a function, or only a part? Part-only changes were logged without discussion.
- Has it been proposed before? Four ideas were flagged and checked against the register during the break.
- Does it require re-homologation? Three ideas were separated into a platform-change list.

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3 Day one - cab structure

3.1 Ideas logged

Item	Idea	Raised by	Mass (kg)	Action
3.1.1	Cab rear panel gauge 1.2 to 1.0 mm with bead stiffening	Cab Structure	11.2	Logged VC-24-0188b
3.1.2	Delete two webs from the door check bracket	Manufacturing	0.84	Logged VC-24-0561
3.1.3	Cab floor gauge reduction with local reinforcement	Cab Structure	7.2	Platform list
3.1.4	Single-piece A-pillar inner in place of two-piece	Manufacturing	2.10	Logged VC-24-0563
3.1.5	Aluminium cab mounting bracket	Cost Engineering	2.84	Duplicate - see 3.2
3.1.6	Commonise LH and RH mirror brackets	Trim	0.62	Duplicate - see 3.2
3.1.7	Hollow door intrusion beam	Cab Structure	3.40	Logged VC-24-0566
3.1.8	Reduce roof panel gauge with acoustic pad deletion	Cost Engineering	4.10	Referred to day two

3.2 Duplicate checks performed during the workshop

Four ideas were flagged in the room as possible duplicates. The programme office checked each against the register during the afternoon break. The results are recorded here because the group asked that the reasoning be minuted.

Item	Flagged against	Outcome	Reasoning
3.1.5	VC-21-0188, implemented FY22	Confirmed duplicate, and reversed	Implemented then reversed in FY24 after field cracking. Conditions for reconsideration at WF-FA/2024/021 section 6 are not met by this proposal.
3.1.6	VC-21-0402, rejected FY21	Confirmed duplicate	Rejected on mirror shake. No change in the design or the measurement since.
3.1.2	VC-23-0219, implemented FY23	NOT a duplicate	VC-23-0219 is the door HINGE bracket. This is the door CHECK bracket, a different part with a different load path.
3.1.9	VC-22-0455, implemented FY22	NOT a duplicate - carry-over	Same change, proposed for the tipper programme rather than the rigid. Logged as a carry-over for duty-cycle assessment.

Chair's comment, minuted at the group's request. The check on item 3.1.2 took eleven minutes and turned on a single word in the component name. The chair observed that this is exactly the check the group cannot reliably perform in the room, and that two of the four flags would probably have been missed without the programme office present.

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4 Day two - trim and interiors

4.1 Ideas logged

Item	Idea	Raised by	Saving (INR/veh)	Action
4.1.1	Headliner substrate to PET felt	Trim	148	Logged VC-24-0502
4.1.2	Single-piece moulded dash carrier	Manufacturing	610	Platform list - tooling lead time
4.1.3	Delete the map pocket moulding from the base variant	Cost	92	Logged VC-24-0571
4.1.4	Common seat frame across base and mid variants	Supplier Dev	340	Logged VC-24-0572
4.1.5	Reduce carpet pile weight in the base variant	Trim	78	Logged VC-24-0574
4.1.6	Door trim clip count 14 to 11	Manufacturing	46	Logged VC-24-0575

4.2 Discussion on item 4.1.1

Field Quality noted that a PET felt headliner had been trialled on an earlier programme and had produced sag complaints in hot climates. Trim responded that the proposed substrate is a higher-density grade. The group agreed to log the idea with the field history attached, rather than either closing it or logging it without the context. Action on Trim to obtain the earlier trial report.

> Action 4.1. Trim to attach the earlier hot-climate trial evidence to VC-24-0502 before committee submission. Due 11 October 2024.

5 NVH package discussion

NVH presented the measured benefit of the door seal and acoustic glass package benchmarked in CVOEM/VAVE/CB-BM/2024/011, in response to item 3.1.8 which proposed deleting the roof acoustic pad.

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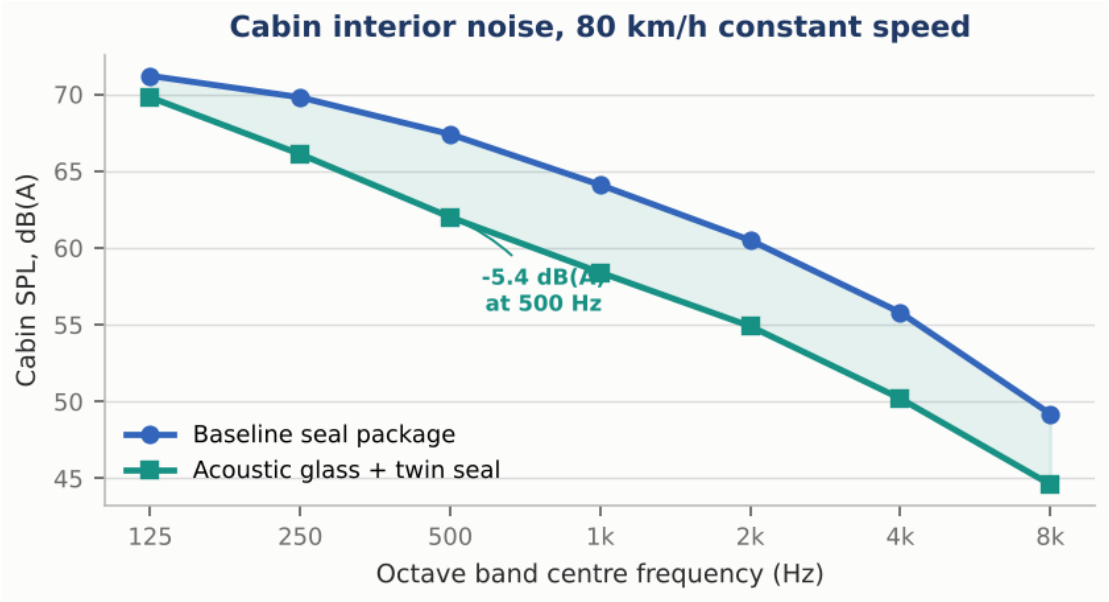


Figure 1 - Cabin interior noise at 80 km/h, baseline seal package against the benchmarked acoustic glass and twin seal package. Presented by NVH on day two.

NVH advised that the roof acoustic pad contributes principally in the 500 Hz to 2 kHz range and that deleting it would cost approximately 2.1 dB(A) at 1 kHz, which exceeds the available margin against the cabin noise target. Item 3.1.8 was withdrawn by the originator.

Outcome 5.1. Item 3.1.8 withdrawn in the room. Minuted with the reasoning so that a future proposal to delete the roof acoustic pad can be closed against this record.

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6 Summary and actions

	Count
Ideas generated	34
Screened out in the room	9
Confirmed duplicates	3
Identified as carry-over	1
Referred to the platform-change list	3
Logged for assessment	25
Withdrawn by originator during the workshop	1

6.1 Actions

Ref	Action	Owner	Due
A-01	Attach hot-climate trial evidence to VC-24-0502	Trim	11 Oct 2024
A-02	Confirm tooling lead time for the dash carrier	Supplier Development	18 Oct 2024
A-03	Obtain homologation view on items 3.1.3 and 4.1.2	Homologation	25 Oct 2024
A-04	Circulate the four duplicate check results to all originators	Programme Office	27 Sep 2024
A-05	Log item 3.1.9 as a carry-over with a duty-cycle note	Programme Office	27 Sep 2024
A-06	Electrical Installation to comment on item 4.2	Electrical Installation	4 Oct 2024

6.2 Chair's closing remark

The chair noted that of 34 ideas generated, four required a check against the existing record and that check could only be made because the programme office attended with access to the register. The chair asked that this observation be minuted and forwarded to the VAVE Committee.

Next workshop: chassis and driveline, provisionally 14 and 15 November 2024.

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